<u>Chemo-Biosensor System</u>

Equipment Description: "Chemo-Biosensor System"-an experimental set-up comprising Conductometric Gas Sensor capable of measuring various biomolecules present in the exhaled Human Breath with the quantities in the order of ppm to ppb level. The system should consist of the following components of high quality.

- **Gas Sensor Chambers** *(Volume: 20 mL or less; one chamber made of SS type 304; 316L or equivalent and other chamber made of Teflon*) with gas inlet/outlet options and provision for 2 separate Sample Holder insertion; Teflon gas chamber should have similar options and, in adduition, must be equipped with UV-light irradiation facility for cleaning the electrode substrate.
- **Sample Holder made of Alumina substra**te (6x3 mm^2) with Pt interdigitated electrodes (Sensors) on one side and a Pt heater on the other side (operable between room temperature and 600°C). The spacing between the Pt electrodes is 200 microns. Sensing material shall be deposited on the Screen Printing Electrodes (SPE) and the modified SPE is to be inserted in the sample holder. Sample holder shall be connected to the the computer for data collection, and power source for heating.
- **Mass flow Controllers** (5 numbers) with thermal mass flow sensor (Bronkhorst High-Tech model F-201CV-100); MFCs for accurate measurement and control of flow ranges (Minimum: 0.8…40 mln/min; Nominal: 0.8…100 mln/min; Maximum: 0.8…150 mln/min) at operating pressures between vacuum and 64 bar with Multi Gas / Multi Range functionality); 2 numbers (Nitrogen and Oxygen) for reference line (carrier gas); 3 numbers (Nitrogen, Oxygen and Mixture) to have the target line with a tunable concentration value of the target gas.

- **Solenoid valves:** total 6 numbers; (2-numbers : Burkert Type 0127 Rocker valve 2/2 or 3/2-way solenoid valve and 4-numbers (Burkert Type 300-C C 3/2-way valve, when de-energized, outlet get exhausted).
- **Interactive Digital Source Meter: Keithley- Model No.2450**
- **Keithley cable**: KUSB-488B IEEE-488.2 USB-TO-GPIB INTERFACE ADAPTER 7007-2 SHIELDED GPIB CABLE: 1 M (3.28 FEET)

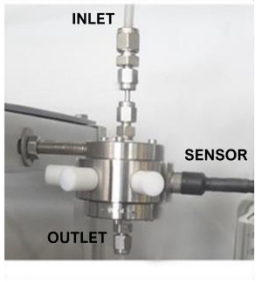
- **Power Source :** TCABUR XCSD30C power supply, a single-phase 30W switching power with small dimensions (71x90x62 mm^3), with class 2 insulation and protection against overload, overvoltage, overheating and short circuit; The power supply of the Solenoid Valve block requires ESP 36-24SN; TRACO POWER. 24 V and a maximum current of 1.5 A or any other equivalent power source with Indian standard input.
- High quality PTFE-TEFLON TUBES for connecting gas bottles, valves, gas sensor chamber, and mass flow controllers.
- High purity gases: N2, H2, O2, Acetone, NH3, NO2, NO, and Ethanol (Permeation tubes filled with the listed chemical compounds and related accessories such as flow meter and constant temperature bath with accuracy ± 0.1 °C)

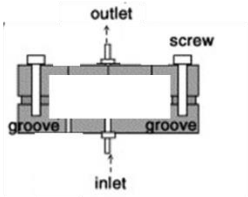
Hardware and software: The System should be equipped with a UDOO x86 Ultra microcomputer board, a microprocessor and a microcontroller compatible with Arduino board specifications.

Software to interface Windows 10 OS, mass-flow controller arrays at the same time with a single GUI (Graphical user interface), and the electrode for simultaneous measuerment of changes in resistance as a function of time, gas concentration and temperature.

<u>Image Dscription of Gas Sensor Chamber made of SS and Teflon</u>

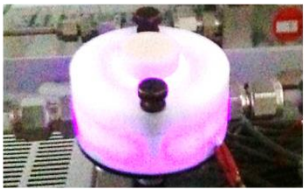
Chamber made of SS

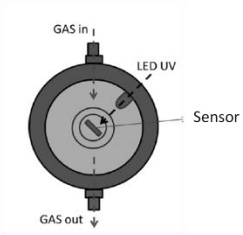


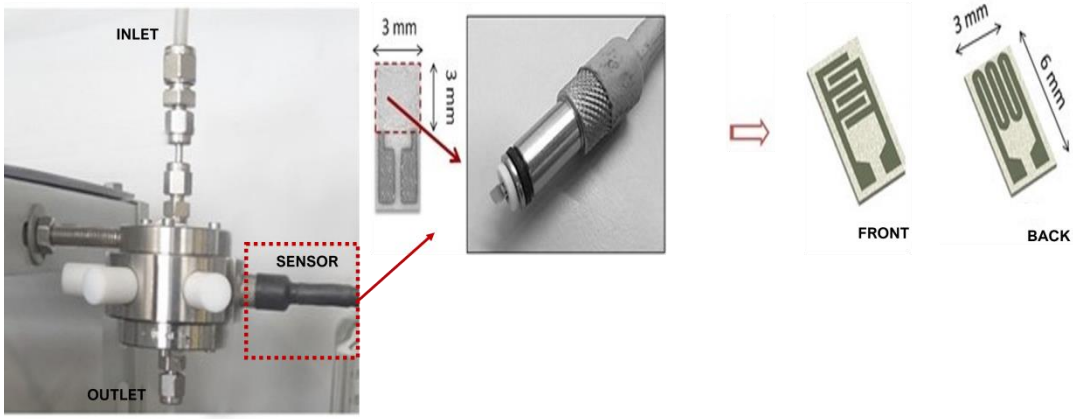


Volume: 20 mL or less;
Can arrange up to six sensors

Chamber made of Teflon





Volume: 10 mL or less;
Equipped with LED for UV-light irradiation of the sensor.

<u>Image Description of Pt interdigitated electrodes (Sensors) with inbuilt Pt heater</u>